HYDROFARMS AGRI SOLUTIONS

GREENHOUSE STEEL STRUCTURE, CONNECTIONS AND RAINWATER SYSTEM

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-STR-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt - open terrain
Prepared by	Hydrofarms Agri Solutions
Issue date	15 August 2026
Parent methods	HF-GH01-MST-GEN-002 and HF-GH01-MST-CIV-002

ERECTION STABILITY IS A TEMPORARY-WORKS DESIGN
No frame shall be left free-standing. Temporary guys and permanent longitudinal ties/X-bracing shall be installed by the approved erection sequence before releasing crane support or exposing the partially erected structure to wind.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor control and approval status.
Construction manager	Erection sequence, lifting coordination and daily work-front release.
Structural engineer	AFC design, temporary stability, connection details and deviation assessment.
Erection engineer	Method implementation, frame plumbing, bolt control and work records.
Surveyor	Base/anchor release, frame coordinates, verticality and final as-built survey.
QA/QC engineer	Material traceability, galvanizing, bolting, inspection and turnover records.
Lifting supervisor	Lift plan, crane configuration, rigging and exclusion zone.
HSE manager	Work at height, wind limits, rescue, hot-work and access controls.
Client/consultant	Hold/witness inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method controls receipt, preassembly, lifting, alignment, bolting, bracing, galvanizing repair, gutter installation and handover of the production and service greenhouse steel structures. It converts the adopted geometry and member coding into a stable erection sequence while protecting civil works and future covering/screen interfaces.

2. Scope and Package Boundary

The steel-structure package includes the following work:

- Production greenhouse columns, arches/roof hoops, bottom chords, longitudinal ties, crown ties and X-bracing.
- Service-greenhouse framing, polycarbonate/covering support interfaces and local opening frames to approved shop drawings.
- Baseplates, anchors, levelling hardware, non-shrink grout, cleats, gussets, splice assemblies, bolts and turnbuckles.
- Internal valley gutters, outlets, downpipes, debris guards, seals and interfaces with roof plastic/polycarbonate.
- Roof-vent structural curbs, hinges, torque-tube/rack supports and temporary closed-position restraints.
- Local independent frames for fans, cooling pads and doors where shown, without transferring equipment vibration to light covering members.

PACKAGE EXCLUSIONS
Concrete foundations belong to the civil/structural foundation package. Fans, pads, screens, covering films, polycarbonate, vent motors, screen motors and electrical cables belong to their respective packages; this steel method includes only their approved structural supports and interfaces.

3. Governing Information and Safety Basis

Priority	Controlled information
1	Signed AFC structural calculations, drawings, connection schedule and foundation reactions
2	Approved shop drawings, piece marks, bolt schedule, weld procedures and erection sequence
3	Approved survey release for foundations, baseplates and anchors
4	Greenhouse design requirements and applicable Egyptian structural/loading provisions
5	UNE-EN 13031-1 greenhouse design/construction requirements where adopted by the designer
6	ISO 1461:2022 galvanizing requirements and approved coating-repair system
7	Supplier data for covering, internal/external screens, roof vents, fans, pads and polycarbonate

DESIGN STATUS
The 20 m/s wind case is the adopted preliminary economy-screening basis, not a substitute for the final signed structural model. Fabrication and erection release require structural-engineer confirmation of project wind, terrain, openings, vents, cladding permeability and suspended loads.

4. Adopted Production-Greenhouse Geometry

Parameter	Adopted value	Erection implication
Overall footprint	126 m long x 36 m wide	Production greenhouse only
Eaves / underside of arches	5.50 m	Bottom chords and eaves ties at this level
Crown level	7.00 m	Crown ties fixed at the true arch apex
Longitudinal bays	14 bays x 9.00 m	15 main frame axes at X = 0, 9, ... 126 m
Secondary roof hoops	14 axes at mid-bays	X = 4.5, 13.5, ... 121.5 m
Width grid	9 arch bays x 4.00 m	10 column/support lines at Y = 0, 4, ... 36 m
Permanent X-bracing planes	X = 0, 45, 81 and 126 m	Eight braced width spans per plane; central 4 m aisle span clear
Service greenhouse	9 x 36 m on west side	Independent coordinated framing and shared-wall interfaces

5. Adopted Member Schedule

Code	Member	Adopted section	Location / function
A1/A2	Main arches and mid-bay hoops	CHS OD101.6 x 4.0 mm HDG	29 roof axes, nine 4 m arch bays
C1	All production columns	RHS 80 x 80 x 4.0 mm HDG	150 columns, 5.50 m high
P1/T1	Longitudinal ties and bottom chords	RHS 50 x 30 x 2.0 mm HDG	Ten longitudinal lines and every roof axis
P2	Crown/ridge ties	CHS OD60.3 x 3.0 mm HDG	Nine continuous lines at Z = 7.00 m
B1	X-bracing	16 mm HDG rods/cables	64 diagonals with turnbuckles and gussets
BP	Baseplates	300 x 300 x 16 mm HDG	One per production column
AN	Anchor bolts	4 x M24 HDG per plate	Double nuts and washers with templates
GT	Internal valley gutters	2.0 mm pre-painted galvanized steel	Eight lines x 126 m, developed width about 490 mm

Steel shall be of the grade stated in the signed design and mill certificates. Section substitutions, reduced wall thickness, drilled site holes in primary members or unapproved welding are prohibited.

6. Core Production-Steel Takeoff

Member	Installed length / qty	Unit mass	Estimated weight
CHS 101.6 x 4 arches/hoops	1,148.40 m	9.628 kg/m	11.057 t
RHS 80 x 80 x 4 columns	825.00 m	9.546 kg/m	7.875 t
RHS 50 x 30 x 2 ties/chords	2,304.00 m	2.386 kg/m	5.498 t
CHS 60.3 x 3 crown ties	1,134.00 m	4.239 kg/m	4.807 t
16 mm bracing rods	435.27 m	1.578 kg/m	0.687 t
300 x 300 x 16 baseplates	150 pc	11.304 kg/pc	1.696 t
Core production structure total	-	-	31.620 t

WEIGHT BOUNDARY
The 31.620 t estimate covers the listed production core members only. It excludes service-greenhouse steel, fan/pad/door/vent local frames, splice/cleat/gusset steel, bolts, gutters, downpipes, screen supports and fabrication allowances. Final procurement weight shall come from the approved fabrication model/MTO.

7. Connection and Accessory Schedule

Item	Installed qty	Purchase qty	Quantity basis
300 x 300 x 16 baseplate	150 pc	153 pc	One per column + 2%
M24 anchors	600 pc	618 pc	Four per plate + 3%
M24 nuts / washers	1,200 / 1,200 pc	1,236 / 1,236 pc	Double arrangement per anchor
Arch spring cleat/gusset set	522 sets	549 sets	29 axes x 9 bays x 2 ends
Arch splice sleeve/plate set	522 sets	549 sets	Two splices per 4 m arch bay allowance
M16 arch connection bolts	4,176 pc	4,385 pc	Four per cleat/splice set
RHS tie end-cleat set	672 sets	706 sets	Longitudinal supports + chord endpoints
M12 tie bolts	1,344 pc	1,412 pc	Two bolts per tie cleat
Crown saddle clamps	261 sets	274 sets	Nine lines x 29 roof axes
M12 crown-clamp bolts	522 pc	549 pc	Two bolts per saddle
16 mm bracing diagonals	64 pc	66 pc	4 planes x 8 spans x 2 diagonals
Turnbuckles	64 pc	66 pc	One per diagonal
Bracing gusset plates	128 pc	135 pc	Two per diagonal

8. Rainwater Schedule

Component	Installed quantity	Purchase quantity	Technical requirement
Internal valley gutter	1,008 m	1,059 m	8 lines x 126 m; 2.0 mm PPGI
Gutter brackets/hangers	680 pc	714 pc	Maximum 1.50 m spacing; closer at outlets/joints
Splice/joint sets	160 sets	168 sets	Based on 6 m pieces; sealed bolted overlaps
Expansion-joint sets	16 sets	17 sets	Two per 126 m gutter line
End caps	16 pc	17 pc	Two per line
Outlet drops + debris guards	64 + 64 pc	68 + 68 pc	Eight outlets per line at about 18 m
Minimum 4-inch downpipes	352 m	370 m	64 drops x 5.50 m
Downpipe clamps	192 pc	202 pc	Three clamps per drop
Butyl/EPDM sealing tape	By joints	30 rolls	Compatible with galvanized/PPGI surfaces
Neutral-cure silicone/PU	By joints	80 tubes	UV-resistant; acidic silicone prohibited

Service-greenhouse gutters, if separate from production gutter lines, shall follow the approved 9 x 36 m shop drawing and be measured once in the final fabrication MTO.

9. Materials, Fabrication and Delivery

- Submit mill certificates, section dimensions, steel grade, galvanizing certificates, bolt certificates and coating-repair data by piece-mark batch.
- Complete cutting, drilling, welding, vent/drain holes and trial assembly before hot-dip galvanizing wherever practicable.
- Fabricate concentric arch splices and bolted cleats so the arch centreline remains continuous without abrupt kinks or local flattening.
- Provide vent/drain holes for closed hollow sections during galvanizing, oriented so water cannot remain trapped after erection.
- Mark every member durably without damaging coating. Pack using padded separators and lift with soft slings; do not drag galvanized members.
- Quarantine bent, twisted, under-thickness, poorly galvanized or incorrectly drilled members pending written disposition.

10. Preconditions

- Signed structural design and erection drawings are released, including temporary-stability and connection requirements.
- Foundation concrete strength, coordinates, levels, anchor projection and access route are accepted under survey hold points.
- Crane standing area and lift radii are verified against ground-bearing capacity and underground tank/pipe exclusions.
- Weather forecast and project wind limits are suitable for each lift; an anemometer is operational at work level.
- Certified crane, lifting accessories, MEWPs/scaffolds, torque tools and rescue equipment are available.
- Fan/pad frames, service greenhouse, roof vents and shading-screen interfaces are coordinated before closing structural lines.

11. Detailed Erection Method

11.1 Base and anchor release

- Survey all 150 production bases for coordinates, top level, anchor spacing, projection and diagonals before steel delivery to the work front.
- Clean threads, fit levelling nuts/washers and establish a common plate datum; do not bend anchors to force plate fit.
- Report out-of-tolerance anchors to the structural engineer. Flame cutting plate holes or enlarging them without approval is prohibited.

11.2 Trial assembly and first stable bay

- Trial-assemble representative 4 m arch segments, splice sleeves and spring cleats at ground level; verify crown height and symmetry.
- Erect the first main frame at X = 0 with all ten columns, arches and bottom chords, using temporary guys in both directions.

- Erect the adjacent mid-bay hoop at X = 4.5 m and main frame at X = 9 m, then install longitudinal ties/crown ties to create the first stable 9 m bay.
- Install permanent bracing assigned to the X = 0 plane before progressing beyond the first bay.

11.3 Progressive frame erection

- Continue in 9 m bays: main frame, mid-bay hoop, next main frame, bottom chords, longitudinal ties and crown ties.
- Maintain temporary guys until the permanent bracing system and sufficient adjacent bays are complete and accepted.
- Check column plumb, frame spacing, spring-line level, crown level and bay diagonals at every three frames and before final bolt tightening.
- Install X-bracing at X = 45 m, 81 m and 126 m as those planes are reached; keep the central 4 m aisle opening clear as detailed.

11.4 Bolting and alignment

- Use the specified bolt grade, diameter, length, nut and washer arrangement. Mixed unidentified bolts are prohibited.
- Snug-tighten during initial assembly, complete survey alignment, then apply the approved final tightening/torque procedure.
- Mark completed bolts for visual inspection and record calibrated-tool identification and tightening batch.
- Tension turnbuckles evenly after global alignment; do not pull frames out of plumb by overtensioning one bracing plane.

11.5 Baseplate grouting

- After frame alignment and structural release, clean and precondition the concrete substrate and form sealed grout shutters.
- Place approved non-shrink grout continuously beneath each 300 x 300 mm plate at nominal 20 mm bedding or AFC detail.
- Ensure full bearing without voids, cure the grout and final-lock anchor nuts only after the specified grout strength is achieved.

11.6 Gutters and downpipes

- Install gutter supports to line and fall at maximum 1.50 m spacing and add supports beside outlets, ends and expansion joints.
- Join 2.0 mm PPGI gutter pieces using the approved overlap plates, EPDM/butyl seals and neutral-cure sealant.
- Provide expansion joints at the scheduled locations, end caps and 64 outlet/debris-guard assemblies.
- Fix minimum 4-inch downpipes with three clamps each and discharge away from foundations, doors, equipment and service circulation.
- Conduct a controlled water test of every line; rectify ponding, leakage and backfall before roof covering interfaces are closed.

11.7 Roof vents and screen interfaces

- Install vent curbs, hinge rails, torque-tube/rack supports and motor brackets to the approved vent shop drawing without distorting arches.
- Maintain vents mechanically locked closed until actuators, limit switches, controls and weather seals are commissioned.
- Install internal/external shading-screen supports and drive brackets only at approved structural nodes; no site holes in primary arches without approval.
- Verify that moving screens, roof vents, gutters, covering lock channels and anti-billow wires have clear independent travel paths.

11.8 Galvanizing repair

- Avoid field welding. Where written approval permits cutting/welding, remove slag and prepare the surface without excessive grinding of parent metal.
- Repair coating by the approved zinc-rich system to the required dry-film thickness and overlap sound galvanizing.
- Record each repair location; extensive coating damage or member distortion requires structural/QA disposition rather than cosmetic repair.

12. Survey and Acceptance Tolerances

Characteristic	Control / acceptance
Frame coordinates	Match AFC grid and accepted foundation control
Column plumb	Within signed structural/erection tolerance over 5.50 m height
Frame spacing	9.00 m main and 4.50 m mid-bay hoop axes without cumulative drift
Width support lines	Y = 0, 4, 8, ... 36 m; no diagonal plan offset
Crown level	7.00 m design level subject to approved erection tolerance
Base bearing	Continuous accepted non-shrink grout, no rocking or unsupported plate edge
Bolts	Correct grade/diameter, complete nuts/washers and approved tightening marks
Gutters	Continuous fall, sealed joints and no ponding/leakage during water test
Coating	Continuous accepted galvanizing/repair with no bare steel or trapped water points

13. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	AFC design/shop drawings/erection sequence	Document review	H	Structural/QA-QC	Erection release
02	Material, bolts and galvanizing delivery	Certificates/visual	H	QA-QC/Consultant	Material release
03	Foundations, anchors and crane access	Survey/visual	H	Civil/Survey/HSE	Base release
04	Ground trial assembly	Dimensional/visual	W	Erection/QA-QC	Trial record
05	First stable bay and temporary bracing	Survey/HSE	H	Structural/HSE/Consultant	Stability release
06	Progressive frame geometry	Survey	W	Survey/Erection	Frame report
07	Permanent X-bracing and ties	Visual/tension	H	Structural/QA-QC	Bracing release
08	Final bolting	Mark/torque records	W	Erection/QA-QC	Bolt report
09	Baseplate grouting	Visual/material test	H	Civil/Structural	Grout release
10	Gutters and downpipes	Level/water test	H	Civil/MEP/Consultant	Rainwater report
11	Vent/screen/covering interfaces	Dimensional/functional	W	Structural/MEP/Covering	Interface release
12	Galvanizing and repairs	Visual/thickness	W	QA-QC/Consultant	Coating report
13	Final as-built and handover	Survey/record review	R	QA-QC/Client	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

14. Hold Points

- HP-ST-01: Signed design, shop drawings and temporary-stability sequence accepted before fabrication/erection.
- HP-ST-02: Foundations, anchors and crane standing area accepted before first lift.
- HP-ST-03: First stable bay, temporary guys and X-bracing accepted before progressive erection.
- HP-ST-04: Global frame alignment and permanent bracing accepted before final bolt tightening/grouting.
- HP-ST-05: Gutters pass level and water tests before roof covering closes their interfaces.
- HP-ST-06: Roof-vent and shading-screen structural interfaces accepted before motors/mechanisms are installed.
- HP-ST-07: Final survey, coating repairs, bolt dossier and rainwater test accepted before covering works start.

15. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Frame instability/wind	Approved sequence, guys/bracing, anemometer and stop-work wind limit	Medium
Work at height	MEWP/scaffold, fall restraint/arrest, rescue plan and controlled access	Medium
Lifting operations	Lift plan, certified crane/rigging, tag lines and exclusion zone	Medium
Dropped objects	Tool tethering, toe boards and exclusion beneath work	Medium
Bolt/wrench injury	Correct tools, gloves and stable work position	Low
Hot work/zinc fumes	Permit, ventilation, respiratory control and fire watch	Medium
Sharp gutters/sheets	Cut-resistant gloves, edge protection and controlled handling	Low
Stormwater/sealants	Collect swarf/waste; prevent uncured sealant and wash water entering soil	Low

16. Records and Handover

- Signed calculations, AFC/shop drawings, temporary-works review and approved RFIs/deviations.
- Mill certificates, section checks, bolt certificates, galvanizing certificates and fabrication traceability.
- Foundation/anchor release, lift plans, daily erection reports and weather/anemometer records.
- Frame/alignment surveys, bracing/turnbuckle records, bolt tightening records and grout reports.
- Gutter/downpipe level and water-test reports, sealant batch records and coating-repair register.
- Vent/screen/fan/pad/covering interface releases, final as-built model/drawings and signed turnover certificate.